East West University

CSE 420 (Computer Graphics)

LAB-5

Implementation of DDA Line Drawing Algorithm

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Objective: To implement the **Digital Differential Analyzer (DDA) line drawing algorithm** for drawing a straight line having a both **positive** and **negative slope**.

DDA (Digital Differential Analyzer) Algorithm

Problem: Write a C / C++ program using **OpenGL (GLUT)** to draw a straight line between two given points $(x1, y1)\ and\ (x2, y2)$ using the **DDA algorithm**, where the **slope of the line is negative**.

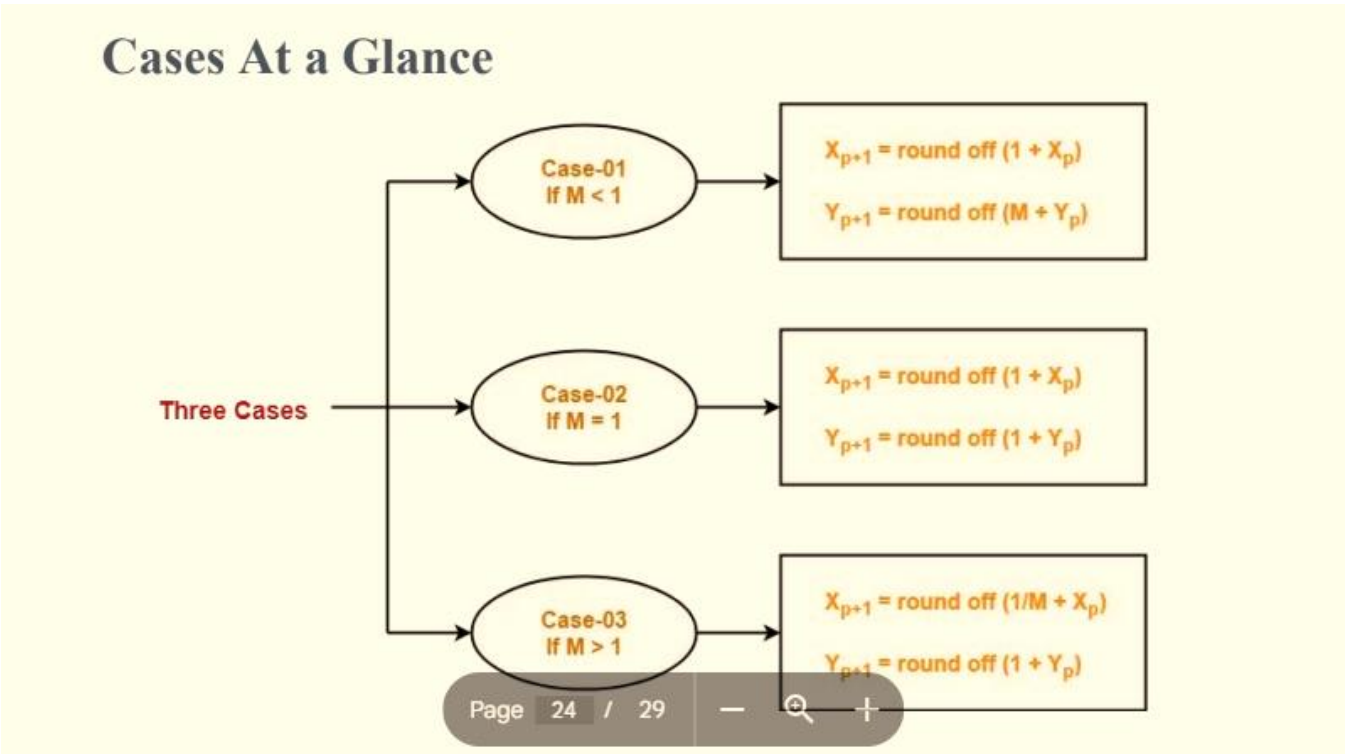

Fig: Cases for DDA Line Drawing Algorithm

Example of DDA Line Drawing Algorithm:

Calculate the points between the starting point (5, 6) and ending point (8, 12).

Solution-

Given-
•Starting coordinates = $(X_0, Y_0) = (5, 6)$
•Ending coordinates = $(X_n, Y_n) = (8, 12)$

Step-01:

Calculate ΔX, ΔY and M from the given input.
•$\Delta X = X_n - X_0 = 8 - 5 = 3$
•$\Delta Y = Y_n - Y_0 = 12 - 6 = 6$

•$M = \Delta Y / \Delta X = 6 / 3 = 2$ => **M>1**

Step-02:

Calculate the number of steps.
As $|\Delta Y| > |\Delta x| = 6 > 3$, so number of steps = ΔY = 6

Step-03:

As M > 1, so case-03 is satisfied.
Now, Step-03 is executed until Step-04 is satisfied.

X_p	Y_p	X_{p+1}	Y_{p+1}	Round off (X_{p+1}, Y_{p+1})
5	6	5.5	7	(6, 7)
		6	8	(6, 8)
		6.5	9	(7, 9)
		7	10	(7, 10)
		7.5	11	(8, 11)
		8	12	(8, 12)

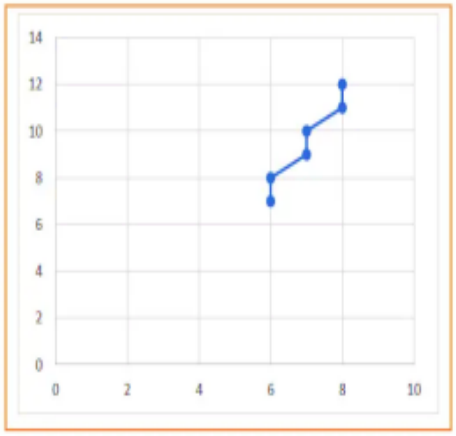

Fig: Example of DDA